

Problems based on Iteration in Python

1. ATM PIN Verification

Problem Statement

An ATM allows a user to enter the correct PIN. The correct PIN is **4589**. The user can keep entering the PIN until it matches the correct one.

Display "**Access Granted**" when the correct PIN is entered.

Sample Output

Enter PIN: 1234

Incorrect PIN

Enter PIN: 9876

Incorrect PIN

Enter PIN: 4589

Access Granted

2. Password Strength Checker

Problem Statement

A website requires users to create a password having at least **8 characters**.

Keep asking the user to enter a password until the entered password satisfies the minimum length requirement.

Sample Output

Enter Password: hello

Password too short.

Enter Password: python@123

Password Accepted.

3. Number Guessing Game

Problem Statement

A secret number is **37**.

Keep asking the user to guess the number until the correct number is entered.

Display whether the entered number is **too high**, **too low**, or **correct**.

4. Multiplication Table Generator

Problem Statement

Write a Python program that accepts a number from the user and displays its multiplication table up to 20.

Sample Output

Enter Number: 8

8 x 1 = 8

8 x 2 = 16

...

8 x 20 = 160

5. Count Prime Numbers in a Range

Problem Statement

Accept two integers representing the starting and ending values of a range.

Display all prime numbers within the range and finally display the total number of prime numbers found.

6. Electricity Bill Analysis

Problem Statement

An electricity department wants to analyze electricity consumption of **N houses**.

Accept the monthly units consumed by each house.

Calculate and display:

- Total units consumed
 - Average units consumed
 - Highest consumption
 - Lowest consumption
-

7. Bank Transaction Summary

Problem Statement

A customer keeps entering transaction amounts.

Positive numbers indicate deposits, while negative numbers indicate withdrawals.

The customer enters **0** to finish.

Display:

- Total Deposit

- Total Withdrawal
 - Final Balance
-

8. Employee Salary Statistics

Problem Statement

A company has **N employees**.

Accept the salary of each employee and determine:

- Highest salary
 - Lowest salary
 - Average salary
 - Number of employees earning more than ₹50,000
-

9. Login System with Maximum Attempts

Problem Statement

A system allows only **three login attempts**.

The correct username is **admin** and the password is **python123**.

If the credentials are correct, display "**Login Successful**".

Otherwise, after three unsuccessful attempts, display "**Account Locked**".

Sample Output

Attempt 1

Username: admin

Password: abc

Invalid Credentials

Attempt 2

Username: admin

Password: python123

Login Successful

10. Student Result Analyzer

Problem Statement

A teacher wants to analyze the marks of **N students**.

Accept the marks of each student (out of 100).

Finally display:

- Highest Marks
 - Lowest Marks
 - Average Marks
 - Number of students who passed (Marks ≥ 40)
 - Number of students who scored distinction (Marks ≥ 75)
-

11. Shopping Cart Billing System

Problem Statement

A supermarket allows a customer to purchase multiple products.

The customer first enters the number of products.

For each product, enter:

- Product Name
- Quantity
- Price per Unit

Finally display:

- Individual Product Cost
- Total Bill Amount
- Most Expensive Product
- Cheapest Product
- Average Product Cost